

Methodology

Team: Bit Root **Event:** The 3 LC Hack AI-thon 2.0

1. Initial Data Analysis and Baseline Modeling

When I started tackling the Kaggle challenge for "The 3 LC Hack AI-thon 2.0," my first step was to analyze the dataset to understand its distribution and characteristics. I decided to train a baseline ResNet18 model for the image classification task because of its solid balance between performance and computational efficiency. Upon completing the initial training phase on the raw dataset, my baseline model achieved an accuracy of 78%.

2. Active Learning and Data Curation with 3LC

To push the model's performance beyond the baseline, I adopted an active learning approach, specifically focusing on the samples where the model showed low confidence.

This is where the power of **3LC (3 Lines of Code)** came into play. By utilizing 3LC's visual selection features—such as the pen tool—I was able to swiftly highlight, inspect, and re-label misclassified or low-confidence predictions. This seamless and rapid data curation cycle allowed me to quickly fix errors and improve the ground-truth quality of my training set.

3. Addressing Class Imbalance

After several iterations of curating the dataset with 3LC, I noticed the model's performance had plateaued. I dug deeper into the training dynamics and loss metrics, which revealed that the model was struggling primarily due to underlying class imbalances within the dataset.

To solve this, I added class weights to the loss function. This critical adjustment meant that the model was penalized much more heavily for misclassifying the minority classes, forcing it to learn a robust representation for all categories rather than favoring the majority classes.

4. Final Model Refinement

Once I implemented the weighted loss function and combined it with the high-quality dataset I had curated using 3LC, the model resumed learning effectively. The synergy between precise, visual data curation and a balanced learning objective allowed my ResNet18 model to improve significantly, ultimately achieving a final accuracy of approximately 95%.